

Moving Response Fibres in Neural Networks: Realizability-Cost Scaling and Transverse Response Amplification

Y.Y.N. Li

Abstract

Neural networks contain large response-preserving parameter spaces, but kernel membership alone does not distinguish inert redundancy from directions that carry adaptive value, nor does it quantify the cost of realizing finite motion as the kernel moves. We model this freedom by the state-dependent response kernel $V_\theta = \ker DR(\theta)$. Prospective CNN and ResNet audits show that selected kernel directions outperform norm- and leakage-matched controls, and that current-kernel transport and source-fixed replay exhibit distinct finite realizability-cost scaling. In a trainable ResNet block, nearly collinear current and source-fixed directions differ by more than three orders of magnitude in active response residual, demonstrating transverse amplification of small fibre drift. A prospective GPT-2 confirmation in a rank-2 LoRA-B subspace extends the scaling separation to disjoint natural-English continuations: all 8/8 new seeds and 32/32 interior nodes pass, with approximately quadratic current-fibre cost and near-linear source-fixed cost. Finally, a restricted six-algorithm CNN audit shows that the path minimizing the declared Moving-Fibre F16 action is not the least expensive path to realize. The results establish a state-dependent moving-fibre mechanism across vision and language-model domains, but not complete-kernel transport, semantic invariance, arbitrary-path minimality or a global variational law.

1 Introduction

Neural networks are assembled from local operations and trained by local updates, yet their acquired adaptability is not transparently attributable to any individual update or parameter. Where does this excess functional complexity reside? The question is therefore geometric: in what structure does the surplus live, and how does a local rule navigate it?

Two established lines bear on this question. Optimization-theoretic accounts explain how overparameterization [18], landscape geometry [10, 14, 8, 5] and implicit bias [16] shape solution selection. Representational and information-geometric accounts explain linearized dynamics [13], parameter-space metrics [2] and feature formation [3]. Together they explain much about why training succeeds and what representations emerge. A third, methodological line — protecting or reusing parameter subspaces during adaptation — is closer to the present setting and is separated from it in Section 1.1.

They leave a specific gap. They do not by themselves distinguish inert redundancy from directions that preserve a declared response while carrying future adaptive value. Nor do endpoint- or loss-based accounts ordinarily separate a path functional used to order response-preserving realizations from the correction cost required to make each finite proposal actually satisfy the declared response. For the low-dimensional responses considered here, overparameterization often makes $\dim \ker DR(\theta) = N - \text{rank } DR(\theta)$ large, but kernel membership alone implies neither adaptive value nor selected motion. This paper measures three things directly: whether selected directions inside the kernel carry adaptive value, whether the kernel itself moves in a way that changes realization cost, and whether a small change of the kernel chart can be strongly amplified in response space. Existing tools contain many of the individual ingredients — null spaces, transported subspaces, pullback metrics and natural gradients — but the present work combines them into a prospective audit of adaptive value, online fibre transport, finite realization cost and restricted path action. Three questions organize what follows: which response-preserving directions carry adaptive

value; how does their admissible space move and become functionally consequential along a path; and, within a declared causal family, do path action and realization cost induce the same ordering?

We therefore take the response fibre as the primary object. For a prospectively frozen external response R and a target output y , the realization fibre and its tangent space are

$$F_y = R^{-1}(y), \quad T_\theta F_y \simeq \ker DR(\theta). \quad (1)$$

Functional freedom is the capacity to change an internal realization while preserving a prospectively declared external response. Its directions are invisible to the response to first order, which is precisely why they are usually dismissed as redundancy; the empirical content of this paper is that selected directions within them need not be inert. All comparisons freeze the declared response and controls prospectively and match first-order response leakage; Section 4 gives the full protocol.

The findings form a three-stage mechanism chain, followed by a strict variational boundary. First, selected response-fibre directions carry adaptive value under prospectively frozen, norm- and leakage-matched controls. This premise is confirmed in the complete final-classifier space of a frozen ResNet-18 representation, without implying that every kernel direction is useful.

Second, the admissible response space moves along a path. In CNN and controlled ResNet constructions, transporting a direction in the current response kernel and replaying a direction frozen at the source produce distinct finite realizability-cost scaling. Third, a trainable ResNet audit shows that minute fibre drift can be amplified by more than three orders of magnitude in active response residual, even when the corresponding parameter directions remain nearly collinear. Together, these results identify a state-dependent moving-fibre mechanism rather than merely a large static null space.

A GPT-2 rank-2 LoRA-B audit then tests the breadth of this mechanism rather than supplying its geometric foundation. After an initial codeword-prompt replication, a prospectively frozen confirmation on disjoint natural-English development and response continuations passes all 8/8 new seeds and 32/32 interior nodes. The GPT correction norm and the CNN Fisher-length proxy are different quantities, so the cross-modal claim concerns the ordering and scaling separation between current-fibre transport and source-fixed replay, not numerical equality of their costs.

Finally, a restricted six-algorithm CNN audit supplies a boundary rather than a fourth coequal mechanism: the path minimizing the declared Moving-Fibre F16 action is systematically not the least expensive path to realize. Moving-fibre geometry is therefore supported, whereas a global variational law remains open.

These results establish causal finite-chart mechanism claims, with different experiments carrying different scopes. The CNN and ResNet constructions establish selected adaptive value, moving-fibre realizability-cost separation and transverse response amplification within their declared charts and radii. GPT-2 provides prospective cross-modal confirmation only within a rank-2 LoRA-B subspace of the final two transformer blocks. The experiments do not show that SGD or Adam naturally follows the transported chart, that a complete response kernel forms a smooth global bundle, that the GPT-2 result extends to the full parameter space or semantic invariance, or that ordinary learning minimizes a global action. Any unifying variational functional must be derived independently rather than fitted to the observed ordering.

Section 2 defines the response-fibre geometry, and Section 3 establishes the functional premise that selected response-preserving directions can carry adaptive value. Sections 4–6 develop the moving-fibre mechanism in CNN and ResNet systems through causal transport, realizability-cost scaling and transverse amplification; Section 5.4 provides a GPT-2 cross-modal confirmation on disjoint natural-English continuations. Sections 7–8 test and delimit the proposed variational interpretation. Sections 9–12 state the claim boundaries, interpretation, limitations and conclusions.

1.1 Relation to existing accounts

Five distinctions fix the object precisely.

Full-function symmetry versus declared response equivalence. Work on parameter-space symmetry [9], permutation re-basining [1] and linear mode connectivity [7] studies transformations that preserve the full function. A response fibre requires only $R(\theta') = R(\theta)$ on a prospectively declared anchor set, so it may contain genuine functional change invisible to that response. Exact function-preserving symmetries are a special subset; their adaptive value is not assumed.

Reparameterization invariance versus a state-dependent kernel. Flatness can be manipulated by reparameterization [4], so all declared parameter-space functionals require covariance checks and pulled-back metrics. The present claim is different: V_θ is indexed by state, and the measured object is how this kernel changes along a realized path rather than how a fixed scalar changes under coordinates.

Image geometry versus kernel geometry. The neural tangent kernel [13] studies the image of a parameters-to-functions linearization. Here the object is the kernel of that linearization at a trained checkpoint within a declared parameter block. Natural-gradient and information-geometric methods [2] supply metric machinery but do not imply that selected null directions carry adaptive value.

Moving distributions versus measured adaptive value. That $\ker DR(\theta)$ varies with θ is standard geometry, not the empirical claim. The test is whether replacing a current response-kernel chart by a source-fixed chart creates a prospectively measurable realization penalty and whether small chart drift is amplified by the current response map.

Null-space protection versus response-fibre navigation. Orthogonal-update and gradient-projection methods [17, 6, 15] protect earlier computations; LoRA [11] constrains updates; task arithmetic [12] reuses learned differences. Here an external response is frozen prospectively, its kernel is recomputed along a path, and adaptive value, transport cost and path action are measured separately. The distinction is response-fibre navigation, not generic null-space protection.

2 Response fibres as functional freedom

Let $R(\theta)$ be an external response map, frozen prospectively on a declared anchor set before any evaluation. If y is a regular value, $R^{-1}(y)$ is a level-set submanifold. More generally, where R has constant rank in a neighbourhood of θ , the constant-rank theorem supplies the corresponding local level-set structure, and $T_\theta F_y = \ker DR(\theta)$. At rank-changing points, only the linearized kernel is used here. At singular points no manifold structure is claimed, and $\ker DR(\theta)$ denotes only the linearized response-preserving space. All audits below operate at trained checkpoints where the required Jacobian actions are evaluated by automatic differentiation, while numerical rank, kernel residuals and finite/JVP consistency are explicitly recorded; the full Jacobian need not be materialized, and the linearized object is well defined whether or not the level set is smooth there. The directions that leave the response unchanged to first order are the response-fibre tangents

$$DR(\theta)v = 0, \quad v \in \ker DR(\theta). \quad (2)$$

For a declared Riemannian or operational metric g on parameter space, the coordinate-covariant projection onto the fibre is

$$P_{\ker DR}^g = I - g^{-1}DR^\top \left(DR g^{-1}DR^\top \right)^\dagger DR. \quad (3)$$

The declared output norm is positive definite on the audited response coordinates; otherwise the operational zero set may strictly contain $\ker DR(\theta)$. The complementary directions are response-normal: motion along them changes the declared response and is subject to ordinary error correction. This split is the working definition of functional freedom. It is relative to R by construction, and nothing below claims an R -independent decomposition.

Two statements must be kept apart throughout. That a direction lies in $\ker DR$ is a structural fact, true of vast numbers of directions in any overparameterized model. That a direction carries adaptive value is a functional fact requiring norm-matched, leakage-matched controls. Section 3 addresses only the second.

3 Adaptive value inside a frozen response fibre

This section establishes the functional premise for everything that follows: motion inside a response fibre is not inert redundancy. Throughout it the fibre is identified once, at a source checkpoint, and held fixed — an assumption Section 4 then removes.

3.1 Frozen response and matched controls

The audits in this section follow the common experimental logic below; the restricted Fisher-action audit in Section 7 follows its own separately frozen decision structure. The discipline is what makes the fibre claim testable at all.

1. **Prospective freezing.** The response map R , the anchor set, the internal representation Q , the metric, the dictionary basis where one is used, and all control arms are fixed before target labels or target gradients are observed. Held-out labels are evaluation-only.
2. **Admissibility before interpretation.** An audit enters its validity domain only if the source task was actually learned above a declared accuracy threshold and if the recorded trajectory can be replayed exactly. Stages failing this are reported as inadmissible and their mechanism outputs are treated as diagnostics, never as evidence (Section 9 and Supplementary Section SA).
3. **Norm and leakage matching.** Every compared arm carries the same tangent norm and retains its own realized response-normal component, and stepwise response-leakage mismatch is measured and gated. Without this, an apparent tangent advantage could simply be a larger step or a hidden response change. The fraction of each realized update that lies in $\ker DR$ is itself a protocol quantity and should be reported alongside any effect, since it bounds how much a tangent intervention can move at all.
4. **Structural controls, not straw men.** The primary control is a deterministic equal-norm random vector in the same frozen kernel. Additional arms include no-tangent motion, sign reversal, shuffled coefficients, random subspaces, normal-only motion, shuffled task descriptors, and development-mean substitutes.
5. **Preregistered effect gates.** Direction cosines, gain-retention fractions, and minimum percentage-point advantages are declared in advance. A pooled significance test does not override a failed magnitude gate.
6. **Joint reporting.** Significance and failed effect-size gates are reported together, in the same sentence where possible. Distinct claims — timescale separation, causal value, prospective prediction — are never pooled into one verdict.

Pooled significance values are not reported for the pooled cross-architecture and cross-task comparisons described below. Those tests pool across architectures and tasks that share a source checkpoint, a frozen dictionary and a response map, so the independent unit and the effective sample size are not the number of comparisons; until the run record states the independent unit, the effective n , the test type, and the handling of zeros and ties, only the effect sizes and the frozen-gate outcomes are reported. The exact small-sample values quoted elsewhere in this paper are computed over architecture-seed runs and are stated with their n .

Unless stated otherwise, the models are a small spatial convolutional network and a tiny vision Transformer, the source domain is MNIST, and the independent target domain is USPS, an independently collected handwritten-digit dataset. Artificial label-bijection tasks share identical inputs and differ only in the target relation, which isolates task identity from input statistics.

3.2 Adaptive value of realized fibre motion

The core positive result: response-invisible directions are not inert redundancy.

Response-normal and response-tangent motion relax on different timescales. At trained checkpoints an exact anchor-logit Jacobian separates the two components, and the projection acts on the realized displacement $\Delta\theta$ rather than on the raw gradient. Both checkpoint formation and continuation preserve optimizer state. Median held-out accuracy was 97.69% (residual network) and 86.11% (Transformer), placing the audit inside its declared validity domain.

Architecture	Normal rate	Tangent rate	Rate ratio	Faster runs	One-sided p
Spatial ResNet	0.023929	0.000197	162.37	6/6	0.015625
Transformer	0.054580	0.021164	2.77	6/6	0.015625

$$\tau_{\text{normal}} < \tau_{\text{tangent}} \quad \text{near the declared trained checkpoints.} \quad (\text{H1})$$

Here τ is an empirically fitted relaxation timescale under matched optimizer-consistent continuation. It is not physical clock time and is not automatically identical to T_{info} ; the reading is interpretive, that different implementation directions require different amounts of optimizer evolution to erase a unit perturbation. Response-preserving motion is the slow mode, which is exactly what makes it a candidate carrier of accumulated structure.

The corresponding causal statement did not close at this stage. Blocking normal displacement cost a median loss penalty of 1.54×10^{-5} in the residual network, below the frozen 10^{-4} threshold; and in the Transformer, blocking tangent displacement did not reduce representation motion in the predicted direction (median penalty -0.00779). Timescale separation and causal value must not be pooled.

Learned tangents beat norm-matched random tangents under a synthetic shift. After source training the anchor Jacobian was frozen, and during adaptation the realized displacement was split into its normal and tangent components. The primary control kept the realized normal component and replaced only the learned tangent by a deterministic equal-norm random vector in the same kernel; a second control removed tangent motion. Parameter redundancy, tangent norm, and first-order response invisibility are thereby held constant across arms.

Architecture	True – random	True – no tangent	Seed wins	Leak mismatch
Spatial ResNet	+12.96 pp	+11.94 pp	8/8 and 8/8	11.85%
Transformer	+18.33 pp	+17.87 pp	8/8 and 8/8	11.47%

All frozen gates passed; across the sixteen paired comparisons the one-sided Wilcoxon result was $p = 1.5259 \times 10^{-5}$. Absolute source, pre-shift and per-arm accuracies are in Supplementary Table S6. Throughout this paper, difference columns report the median of paired per-seed differences, not the difference of the two medians; the two estimators need not coincide and the former is the quantity the paired tests are run on.

$$\Delta\theta_{\text{learned}}^{\parallel} \text{ contributes more shifted-domain adaptation than an equal-norm random direction in } \ker DR. \quad (\text{H2-F})$$

The advantage survives on an independently collected domain. The same comparison was run outside the synthetic shift: models trained on MNIST, adapted to USPS, response frozen on twelve MNIST anchors, and for each architecture and each of four seeds the learned tangent compared with eight deterministic equal-norm random kernel vectors and a no-tangent arm.

Architecture	True – random	True – no tangent	Consistent seeds	Median / p95 leak mismatch
Spatial CNN	+7.33 pp	+7.19 pp	4/4	17.44% / 19.40%
Tiny ViT	+8.99 pp	+8.86 pp	4/4	10.08% / 14.61%

All frozen gates passed and the pooled one-sided Wilcoxon p -value was 0.00390625. Absolute accuracies are in Supplementary Table S6.

$$A_{\text{USPS}}(\Delta\theta_{\text{learned}}^{\parallel}) > \text{median} A_{\text{USPS}}(\Delta\theta_{\text{random}}^{\parallel}) \quad (\text{H2-ID})$$

This is the paper’s most robust claim, and its boundary is equally definite. The comparison is conditional on the declared architectures, source anchors, optimizer, target adaptation set, and stepwise leakage gates. It does not show that every response fibre carries semantic capacity, that the result survives natural distribution shift or large models, or that the learned direction minimizes any variational functional.

3.3 Task conditioning and limits

Tangent value depends on task identity and on sign. Three frozen label-bijection tasks used identical inputs but distinct target relations. Each target retained its own response-normal sequence while tangent content was exchanged across tasks, with exactly matched tangent norms and median leakage mismatch below 12%.

Quantity	Spatial CNN	Tiny ViT
Task diagonal advantage	+34.39 pp	+21.63 pp
Own tangent – random	+32.44 pp	+19.53 pp
Own tangent – sign reversal	+45.89 pp	+24.86 pp
Diagonal winners	18/18	18/18
Correct-gradient Spearman	0.671	0.359
Shuffled-gradient Spearman	−0.167	−0.006

The total effect was large and fully consistent. The complete mediation claim did not pass, because the ViT correct-gradient correlation fell below the frozen 0.50 gate. Permitted: tangent value depends on task identity and direction sign. Prohibited: raw Euclidean gradient alignment universally selects the tangent.

The shared correction is a coarse prior, not a universal direction. The correction learned on affine label bijections was frozen and evaluated on an independently specified family of non-affine tasks, with identical source-local geometry, replay state, normal recovery, oracle norms, and evaluation budget across arms, against geometry-only, sign-flipped, and seven norm-matched random corrections:

$$u_{\text{shared}}(t) = \text{Normalize}[u_{\text{geo}}(t) + \alpha \delta u_{\text{shared}}(t)] \quad (4)$$

Quantity	Spatial CNN	Tiny ViT
Shared oracle-gain retention (% of oracle gain)	95.7%	84.8%
Shared – geometry gain	+1.17 pp	+2.97 pp
Shared – random-median gain	+4.67 pp	0.00 pp
Random-direction win fraction	85.7%	42.9%
Median response-leak mismatch	0.020	0.016
Exact replay error	0	0

Coefficient cosines and the own-condition and sign-flipped arms are in Supplementary Table S5. The protocol was admissible in both architectures. The CNN result is substantial evidence for a reusable correction prior: nearly all oracle gain retained, clear margins over sign-flipped and random controls. The ViT result is weaker — much oracle gain retained and noninferior to the task-conditioned arm, but not directionally precise and no median advantage over random corrections.

Pooling architectures and tasks gave positive one-sided orderings against geometry-only motion, sign reversal, and the random-control median. These population orderings do not override failed architecture-specific magnitude gates: median cosine missed 0.75 in both architectures, the gain over geometry-only motion missed 3 points in both, and the ViT failed both random-control gates. The cross-family shared-correction claim is not supported as stated.

Some useful structure does survive transfer, so the correction is not mere within-family memorization. But the transferred object is better described as a coarse, architecture-dependent prior than as a universal functional direction.

3.4 Out-of-construction confirmation in a frozen ResNet-18 representation

Frozen scope. A torchvision ResNet-18 with ImageNet-1K pretrained weights, backbone entirely frozen, on CIFAR-10. The response and the fibre are declared on the complete 5,130-parameter final classifier, so no subchart approximation enters. Sixteen new seeds, 64726–64741, at 4, 16 and 32 anchors, with 16 anchors as the primary gate; 32 matched random kernel directions per seed; confirmation labels used for evaluation only; float64 Jacobian, SVD and finite-step verification.

Frozen gates. At least 12/16 seeds at the primary anchor count and at least 10/16 at the sensitivity counts, with per-seed thresholds $\Delta L_{\text{true}} \leq -0.001$, $\Delta L_{\text{anti}} - \Delta L_{\text{true}} \geq 0.002$, $\Delta L_{\text{random,best}} - \Delta L_{\text{true}} \geq 0.001$ and $\Delta L_{\text{shuffled}} - \Delta L_{\text{true}} \geq 0.001$. Numerical eligibility required $\|Jv\|_2 / (\|J\|_2 \|v\|_2) \leq 10^{-12}$ and $\|\Delta R\|_\infty \leq 10^{-9}$.

Quantity	4 anchors	16 anchors	32 anchors
Candidate seeds	16/16	16/16	16/16
Projected-gradient share, median	0.453	0.338	0.282
True loss change, median	−0.02705	−0.01842	−0.01450
Anti-true margin, median	0.06428	0.04149	0.03162
Best-random-true margin, median	0.02616	0.01784	0.01380
Shuffled-true margin, median	0.02624	0.01742	0.01374
Max kernel residual	3.70e−14	2.27e−13	4.22e−13
Max finite response leakage	5.51e−14	2.98e−13	5.06e−13

Every seed passed every frozen gate at all three anchor counts. Increasing the response declaration reduced the median projected-gradient share from 0.453 to 0.282 and reduced, but did not eliminate, the adaptive advantage. The result therefore confirms the ordering of selected response-fibre tangents over matched controls across anchor-set size; it does not establish anchor-independent effect magnitude.

The backbone was not adapted and the response kernel was not transported. This confirms selected tangent value outside the original construction; it does not replicate the moving-fibre or realizability-cost results of Sections 4–8.

3.5 Prospective GPT-2 LoRA confirmation

Frozen scope. A frozen GPT-2 backbone with rank-2 LoRA adapters. The response and the fibre are declared on the LoRA-B parameters of the final two transformer layers, on built-in codeword prompt tasks. Eight new seeds; sixteen matched random kernel directions per seed; confirmation labels used for evaluation only.

Result. The selected response-kernel tangent outperformed the no-tangent, sign-reversed, shuffled-target and best-of-sixteen random-kernel controls in 8/8 seeds. Every frozen functional, effect-size and response-preservation gate passed.

This audit extends the adaptive-value premise across modality. It concerns a rank-2 LoRA-B block in the last two layers under built-in codeword prompts, not the complete GPT-2 parameter space or general language understanding. This Section 3.5 audit freezes the kernel and tests no moving path. Section 5.4 reports both a constructed current-versus-fixed codeword audit and its prospective confirmation on disjoint natural-English continuations. Together, the Section 3 results establish only the functional premise for what follows: selected response-fibre motion need not be inert.

4 Causal navigation of moving response fibres

4.1 From a source kernel to a moving distribution

Every audit in Section 3 freezes $V_\theta = \ker DR(\theta)$ at a source checkpoint and works inside it. The construction is legitimate, and Section 2 states it as a first-order object at a point. But nothing in that construction says the fibre stays put. As parameters move, $DR(\theta)$ changes, and with it the space of response-preserving directions. Two readings are then possible and they are not equivalent.

Static reading. The response fibre is one subspace. Learning navigates inside it, and identifying it at the source point loses nothing that matters over the distances actually travelled.

Moving reading. The response fibre is a distribution V_θ varying with θ . Motion that is response-preserving at the source becomes progressively less so as the path advances, and a stale fibre is a decreasingly good description of what the network is actually free to do.

Navigation is used operationally throughout, not rhetorically: it means generating each admissible tangent proposal from current and past information while maintaining membership in the current response kernel. Writing H_{static} for $V_{\theta_k} \approx V_{\theta_0}$ and H_{moving} for $V_{\theta_k} \neq V_{\theta_0}$, the two readings make a testable prediction that differs in kind, not in degree. Under the static reading, replaying a trajectory against the source fibre and against the current fibre should cost the same up to noise. Under the moving reading, the two should diverge systematically with distance travelled, and the divergence should have a measurable scaling.

4.2 Causal online navigation protocol

Both arms begin from the same trained source model and receive the same ordered adaptation minibatches, response anchors, metric probes, capability endpoint, Fisher step radius, Newton retraction rule and computational budget. They differ only in the chart used to generate each tangent proposal. They are therefore two online paths matched in source state, data order, response target, step-radius schedule, capability endpoint and budget — not two realizations of one parameter displacement, and not guaranteed to share a parameter endpoint or a total displacement.

Moving arm. The chart is transported and re-projected into the current kernel at each step,

$$B_k = \text{TransportProject}(B_{k-1}, \ker DR(\theta_k)), \quad \|DR(\theta_k) B_k\| \leq \epsilon_{\ker}, \quad (5)$$

with $B_k \in \mathbb{R}^{N \times 8}$. The construction is causal: it uses current and past information only, and reads no future minibatch, no future Jacobian and no final result. It is a current-kernel transported chart, not a dynamically solved 4146-dimensional kernel.

Fixed arm. $B_k = B_0$ throughout, but every step still passes through the same Newton response retraction. The fixed arm is therefore not a deliberately inadmissible straw man: it is a stale tangent space combined with an identical constraint-maintenance mechanism.

Realizability cost. A trial step is taken in the chart and then retracted onto the frozen response,

$$\theta_{k+1}^{\text{trial}} = \theta_k + B_k \Delta z_k, \quad \theta_{k+1} = \text{Retr}_{R(\theta_0)}(\theta_{k+1}^{\text{trial}}), \quad (6)$$

the retraction being a Newton response correction. The single-step realizability cost is the local output divergence the retraction had to remove, and the path cost is its sum:

$$c_k = \sqrt{2D_{\text{KL}}(p_{\theta_{k+1}^{\text{trial}}} \| p_{\theta_{k+1}})}, \quad C(h) = \sum_k c_k. \quad (7)$$

C measures the local output deviation that must be removed in order to turn a proposed tangent step into an actual response-preserving one; a chart that is exactly right incurs none of it. The divergence is measured on the frozen metric probes, so C is a retraction Fisher-length proxy — not a physical energy, a wall-clock duration, or the only possible notion of realizability cost.

Step radii and fitting. Each arm is run at four step sizes, $h \in \{0.08, 0.04, 0.02, 0.01\}$, and a single exponent is fitted per seed by ordinary least squares on

$$\log C(h) = \alpha \log h + \beta. \quad (8)$$

Exponents are reported as medians over comparable seeds, with fit quality alongside. The moving arm predicts $\alpha_{\text{moving}} > 0$; the fixed arm predicts $\alpha_{\text{fixed}} \approx 0$.

5 Confirmed scaling of realizability cost

5.1 Frozen hypotheses and gates

Frozen gates. Eligibility: source accuracy ≥ 0.90 ; source-chart kernel residual $\leq 10^{-7}$; Fisher metric certification passed; all eight paths admissible; at least 14/16 seeds fully comparable. Per path: response-retraction relative error ≤ 0.002 ; retraction-to-step ratio ≤ 0.20 ; moving-kernel residual $\leq 10^{-5}$; effective metric rank ≥ 8 ; maximum step principal angle ≤ 0.80 rad; action left-versus-trapezoid relative change ≤ 0.08 . Scaling: $\alpha_{\text{moving}} \geq 0.50$ with $R_{\text{moving}}^2 \geq 0.80$; $|\alpha_{\text{fixed}}| \leq 0.25$ with relative span $(\max C_{\text{fixed}} - \min C_{\text{fixed}})/\text{median } C_{\text{fixed}} \leq 0.20$; separation $\alpha_{\text{moving}} - \alpha_{\text{fixed}} \geq 0.25$; cost ratio $C_{\text{fixed}}/C_{\text{moving}} \geq 2$ and residual ratio $r_{\text{fixed}}/r_{\text{moving}} \geq 2$. At least 12 seeds must pass every substantive gate. Full per-gate outcomes are in the supplementary gate table.

One seed was excluded: seed 70741, whose source accuracy was 0.898 against the frozen 0.900. Its metric and chart certifications passed; it failed source-model eligibility and was excluded under the frozen rule before any comparison.

5.2 Prospective 16-seed confirmation

Fifteen of sixteen seeds retained full comparability, and all fifteen passed every preregistered gate.

Quantity	Value	Unit
Fully comparable seeds	15/16	count
Seeds passing all frozen gates	15/15	count
Median moving exponent α_{moving}	0.684	dimensionless
Moving exponent range	0.583–0.863	dimensionless
Median fixed exponent α_{fixed}	0.00715	dimensionless
Fixed exponent range	−0.0124–0.0489	dimensionless
Median moving R^2	0.995	dimensionless
Minimum moving R^2	0.976	dimensionless
Median fixed relative span	3.05%	relative
Maximum fixed relative span	11.16%	relative
Median cost ratio	21.30	dimensionless
Smallest-radius cost-ratio range	13.61–26.17	dimensionless
Tangent residual ratio	21.47	dimensionless
Excluded seed 70741 source acc.	0.898	< 0.900

The separation is not carried by a few outlying seeds. The moving exponent stays between 0.583 and 0.863 with median $R^2 = 0.995$ and minimum 0.976, so the power law is well determined in every comparable seed; the fixed-arm exponent was close to zero under the prospectively frozen absolute-exponent and relative-span gates, with median 0.00715 and range −0.0124 to 0.0489, and a median relative span of 3.05% across the four radii. The cost ratio at the smallest radius ranges from 13.61 to 26.17.

Post-confirmation robustness analysis. This analysis was not a preregistered gate. Leave-one-radius-out refits changed the median moving exponent by at most 0.048 and the median fixed exponent by at most 0.003. The smallest separation between the two median exponents across the four refits was 0.627, and the smallest separation over all 60 individual seed-radius refits was 0.545; both remain above the frozen 0.25 gate. The per-seed table is in the Supplementary Information, and the aggregate table and its reconstruction script are in the public archive; the whole analysis is derived from the archived per-seed records.

The two ratios agree closely with one another (21.30 and 21.47). Their agreement is consistent with a common geometric source, but does not identify that source or establish a causal relation with fibre rotation.

5.3 Cross-model confirmation in ResNet-18 / CIFAR-10

The cost-scaling separation is not confined to the original construction. A prospective sixteen-seed audit used a frozen ImageNet-pretrained ResNet-18 backbone with a trainable terminal residual adapter and classifier, and applied the same current-kernel transport and source-fixed replay comparison.

Quantity	Value
Seeds passing every geometric and held-out gate	16/16
Moving-minus-fixed slope separation, median	0.575
Smallest-radius fixed / moving cost ratio, median	2.997
Raw development / confirmation gradient cosine	0.414
Projected cosine	0.996

The permitted conclusion is that the realizability-cost scaling difference between current-kernel transport and source-fixed replay is not restricted to TinyCNN/MNIST. It is not a universal scaling law: the result remains specific to this ResNet construction, these finite radii and this transported chart.

5.4 GPT-2 LoRA current-versus-fixed replication

An analogous current-kernel versus source-fixed comparison in frozen GPT-2 rank-2 LoRA-B first passed 6/6 seeds and 18/18 codeword-prompt nodes, then passed a prospective confirmation on disjoint frozen natural-English development and response sets in 8/8 new seeds and 32/32 nodes. At state φ , with $J_\varphi = DR(\varphi)$, response error $e_h^{(a)} = R(\varphi + hv^{(a)}) - R(\varphi)$ was assigned the regularized Newton correction

$$\delta\varphi_{N,h}^{(a)} = -J_\varphi^\top \left(J_\varphi J_\varphi^\top + \lambda I \right)^{-1} e_h^{(a)}, \quad C_{\text{GPT}}^{(a)}(h) = \left\| \delta\varphi_{N,h}^{(a)} \right\|_2. \quad (9)$$

C_{GPT} is a correction norm in the declared LoRA-B coordinates, not the CNN Fisher-length proxy.

Quantity	Range
Seeds / interior nodes passing	8/8; 32/32
$\alpha_{\text{current}} / \alpha_{\text{fixed}}$	1.9987–2.0040 / 1.0625–1.2106
Finest-radius $C_{\text{fixed}}/C_{\text{current}}$	8.87–35.56
Current-minus-fixed slope gap	0.792–0.939
Active-Jacobian residual ratio	1.41e7–1.12e8
Max post-retraction response error	1.08e–15

Re-projection makes first-order invisibility of the current arm true by construction. The non-trivial result is the stable finite normal component and amplification acquired by source-fixed replay. The V3 confirmation shows that this separation is not confined to the original codeword prompts: it persists on frozen, disjoint natural-English continuations under all prospective gates. This supports cross-modal moving-fibre mechanism replication and rejects a single fixed source chart over the tested radii. It does not establish semantic invariance, natural training navigation, full-parameter geometry, F16 ordering, Euler–Lagrange stationarity or a global variational law.

6 Transverse response amplification in a trainable ResNet block

A moving fibre raises a question the cost measurements do not answer: how far must the fibre turn before the difference matters? The answer is that it need not turn far at all.

6.1 Active directions

At a point on the path, let B_m be the transported and re-projected current chart and B_0 the source-fixed chart. Both have Euclidean-orthonormal columns after transport and re-projection, so $B_m B_m^\top$ and $B_0 B_0^\top$ are the corresponding chart projectors. The two descent proposals inside them are

$$d_m = -\frac{B_m B_m^\top \nabla L}{\|B_m B_m^\top \nabla L\|}, \quad d_f = -\frac{B_0 B_0^\top \nabla L}{\|B_0 B_0^\top \nabla L\|}, \quad \delta d = d_f - d_m. \quad (10)$$

6.2 Transverse amplification

Here $J = DR(\theta_k)$ is the Jacobian of the prospectively frozen response map evaluated at the current path state. The response declaration is frozen; the Jacobian value is not — this is exactly what separates $DR(\theta_k)$ from $DR(\theta_0)$. Two distinct ratios must be kept apart. The active residual ratio compares how visible the two proposals are to the frozen response,

$$\|Jd_f\| / \|Jd_m\|, \quad (11)$$

while the transverse gain contrast compares the response gain along the difference direction with the gain along the current-fibre direction,

$$A_{\perp} = \frac{\|J\delta d\|}{\|\delta d\|}, \quad \Gamma_{\perp} = \frac{\|J\delta d\| / \|\delta d\|}{\|Jd_m\| / \|d_m\|}. \quad (12)$$

These are not the same quantity and are never reported as one.

6.3 Prospective confirmation

A sixteen-seed prospective audit on a trainable ResNet-18 layer4-plus-classifier system, using an eight-dimensional matrix-free transported response-kernel chart, passed with 16/16 against a frozen 12/16 gate.

Quantity	Median
Seeds passing (frozen gate 12/16)	16/16
Direction cosine between d_m and d_f	0.999990
Active residual ratio $\ Jd_f\ /\ Jd_m\ $	1437.844
Transverse response gain $A_{\perp}$	90.108
Transverse gain contrast $\Gamma_{\perp}$	341,763
Finest finite-difference / JVP ratio	0.999998
JVP additivity error	5.05e−14

Two parameter directions that are almost collinear need not be functionally equivalent in the current response geometry: a minute angular drift of the fibre can fall into a high-gain transverse direction. The finite-difference agreement shows this is a property of the response map, not a linearization artifact.

6.4 Precision certification and its failure history

The signal gates of the first attempt passed, but its JVP identity certification failed under float32 with TF32 matrix multiplication enabled. The repair ran in float64 with TF32 disabled and passed development certification; the confirmation above then used a fresh set of sixteen seeds. The precision failure was caught by a preregistered certification gate, and no result from the failed stage is reported as positive.

7 Confirmed Moving-Fibre F16 ordering

Once the fibre is known to move, a natural follow-up is whether some functional selects a path within it. The direction of inference matters: the moving-fibre geometry of Sections 4–5 suggests a candidate path-selection functional, not the reverse.

7.1 Frozen question

The question is whether the natural-gradient path minimizes a declared action among a fixed set A_6 of six causal online algorithms, at every tested step radius:

$$\gamma_{\text{nat}} \in \arg \min_{\gamma \in A_6} S_{\text{MF16}}[\gamma; h], \quad h \in \{0.08, 0.04, 0.02, 0.01\}. \quad (13)$$

The action is a discrete functional on the realized sequence. At step k , with transported chart B_k , let $G_k = \text{regularize}(B_k^{\top} F_{\theta_k} B_k)$ be the eigenvalue-floored output-Fisher tensor in chart coordinates and let

$g_k = B_k^\top \nabla L$ be the capability gradient in the same coordinates. The capacity factor is

$$H_k = h_0 + \frac{\sqrt{g_k^\top G_k^{-1} g_k}}{s_0}, \quad h_0 = 0.1, \quad (14)$$

with s_0 the same quantity evaluated at the source point, so that H_k is dimensionless and equals $h_0 + 1$ at the source. For the realized increment $\Delta\theta_k = \theta_{k+1} - \theta_k$, write its chart coordinates at the two endpoints as $z_k^- = B_k^\top \Delta\theta_k$ and $z_k^+ = B_{k+1}^\top \Delta\theta_k$, and the corresponding Fisher lengths as $d\ell_k^- = \sqrt{z_k^{-\top} G_k z_k^-}$ and $d\ell_k^+ = \sqrt{z_k^{+\top} G_{k+1} z_k^+}$. The action is the trapezoidal sum

$$S_{\text{MF16}}[\gamma, h] = \sum_{k=0}^{K-1} \frac{1}{2} \left(\frac{d\ell_k^-}{H_k} + \frac{d\ell_k^+}{H_{k+1}} \right). \quad (15)$$

Both the metric and the capacity are therefore re-evaluated at every node, and each increment is measured in the chart at both of its endpoints. S_{MF16} carries the units of Fisher length; H is dimensionless by the source normalization. The step radius h is the algorithmic Fisher step radius, not an integration mesh: the nodes are the realized steps, and K is set by first arrival at the capability endpoint. The frozen convergence statistic is the relative difference between this trapezoidal sum and the corresponding left-rule sum $\sum_k d\ell_k^- / H_k$, gated at 0.08. The wrong-metric arm replaces G_k by $\Pi^\top G_k \Pi$ with Π the reversal of the chart coordinates, and recomputes H accordingly. The pure Fisher length used in Section 7 is the same sum without the H divisors.

I The exact serialized implementation, including the left and trapezoidal evaluations and the frozen convergence statistic, is in Supplementary Item 13 and the public archive.

Every algorithm is generated online and causally; recomputes the current response kernel at each step; uses the transported eight-dimensional chart of Section 4.2; reaches the common 20% capability-loss-reduction endpoint; and shares the data order, metric protocol and retraction rule. The construction is therefore the same moving-current-kernel construction as Section 5, applied to a path-selection rather than a cost-scaling question.

7.2 Frozen confirmation gates

The core preregistered gates are: at least 14/16 seeds fully comparable; natural gradient winning at all four radii in at least 12/16; beating the wrong-Fisher arm in at least 12/16; positive cost scaling for all six algorithms; fine-radius action convergence for all six; and natural gradient winning under the wrong metric in at most 2/16. Full thresholds are in the supplementary gate table.

7.3 Prospective result

Fourteen of sixteen seeds retained full comparability, and all fourteen eligible seeds passed. In every one of the fourteen, true natural gradient attained the lowest Moving-Fibre F16 action at every tested radius.

Quantity	Value
Seeds attempted / fully comparable	16 / 14
Natural gradient is the action minimizer at all four radii	14/14
Natural gradient beats the wrong-Fisher arm	14/14
Natural gradient wins under the wrong metric	0/14
Median natural cost exponent $\tilde{\alpha}_{\text{nat}}$	0.716
Median fit quality $\tilde{R}^2$	0.993
Median action advantage over the best competing algorithm ($h = 0.08$)	22.8%
Same, seed range	6.3%–41.1%

The same winner is obtained from pure moving-Fisher length, so the present experiment confirms the restricted ordering but does not isolate an independent contribution from the capacity normalization.

The conclusion is an $\arg \min$ statement over the frozen six-algorithm family A_6 only; no claim is made about minimization over arbitrary causal paths.

Seeds 73730 and 73732 were excluded because the coarse-radius natural arm exceeded the frozen `wrong_left_trapezoid_relative_change` threshold, at 0.10676 and 0.10358 against 0.08. Their source, metric, kernel, endpoint and response-retraction diagnostics otherwise remained regular; the frozen exclusion was retained without post-hoc relaxation. The result is therefore 14/16 eligible, 14/14 eligible seeds passed — not 14/14 of the seeds attempted.

8 Restricted action minimization is not realizability-cost minimization

The strongest counterintuitive result is not merely that the response fibre moves, but that the functional selecting a path and the cost of realizing that path produce systematically different orderings.

Section 5 measures what it costs to realize a proposed step; Section 7 measures which path minimizes a declared action. If the action were simply tracking realizability, the same arm would win both. It does not:

$$\gamma_{\text{nat}} \in \arg \min_{\gamma \in A_6} S_{\text{MF16}}[\gamma] \quad \text{while} \quad \gamma_{\text{nat}} \notin \arg \min_{\gamma \in A_6} C_{\text{real}}[\gamma]. \quad (16)$$

The gap is not marginal. At the smallest radius,

$$C_{\text{real}}^{\text{nat}}(h_{\min}) / \min_a C_{\text{real}}^a(h_{\min}) = 2.25 \text{ (median),} \quad \text{range 1.60–4.19,} \quad (17)$$

and the ratio stayed above 1.5 in all fourteen comparable seeds. The action-minimizing path is systematically more expensive to realize than the cheapest admissible alternative, by a factor that never falls to unity.

The separation does not come from the capacity normalization. A natural objection is that both compared quantities are declared here, so their disagreement could be an artifact of the capacity factor H in Equation (14). It is not. Removing H entirely and ranking the arms by pure moving-Fisher length gives the same winner as the action in 14/14 comparable seeds, and that winner is again natural gradient in 14/14. The separation therefore holds between the moving-Fisher length — a standard information-geometric object that this paper does not define — and the retraction cost. It does not depend on h_0 , on the source normalization, or on the capability probe.

This is a result, not a defect of either measurement. The safe form of the claim keeps every qualifier: the experiment rejects the identification of the declared restricted path action with raw realizability cost in the audited system — six declared algorithms, four frozen radii, one architecture and dataset. It does not show that path-selection functionals differ from costs in general, that optimizers actively pay a premium, or that any two-objective principle governs neural training. Nor does it contradict analytical mechanics, where an action was never required to equal an energy or a cost; what is falsified is an order-preserving identification that is easy to assume in a learning setting.

Two consequences follow for the variational line. Any continuum theory must derive a combination of the two quantities from first principles, retain a genuinely multi-objective structure, or replace the vanishing discrete cost by a renormalized density. And the open theoretical question is no longer only whether some functional is globally minimized, but why the lowest-action path is willing to pay a higher realization correction.

9 Negative results and claim boundaries

Claim	Status	Decisive evidence	Boundary
Response-fibre tangents carry adaptive value	Confirmed	Learned tangents beat equal-norm random and no-tangent controls on an independent domain, 8/8 runs, leakage matched	CNN / ViT, frozen finite audits
Selected response-fibre tangent value, out of construction	Confirmed outside the original construction	16/16 seeds pass every gate at 4, 16 and 32 anchors against sign-reversed, shuffled-target and best-of-32 random-kernel controls	Frozen ResNet-18 representation, CIFAR-10, complete final classifier
GPT-2 LoRA selected tangent value	Confirmed	8/8 seeds against no-tangent, sign-reversed, shuffled-target and best-of-16 random-kernel controls	Rank-2 LoRA-B of the last two layers, built-in codeword prompts
GPT-2 LoRA natural-text moving-fibre confirmation	Supported	8/8 seeds, 32/32 nodes; α_{current} 1.9987–2.0040; α_{fixed} 1.0625–1.2106; cost ratio 8.87–35.56; slope gap 0.792–0.939	Rank-2 LoRA-B; disjoint frozen natural-English sets; current direction re-projected at every node
Moving-fibre cost scaling	Confirmed in CNN and ResNet constructions	CNN–MNIST 15/16 eligible, 15/15 passed, $\alpha = 0.684$ against 0.007, cost ratio 21.30; ResNet-18/CIFAR-10 16/16, slope separation 0.575, cost ratio 2.997	Finite charts and radii; not a universal scaling law
Transverse response amplification	Confirmed	16/16 against a frozen 12/16 gate; direction cosine 0.999990 with active residual ratio 1437.8 and gain contrast $3.4e5$; finite/JVP ratio 0.999998	Trainable layer4 + fc, eight-dimensional chart
Complete response-kernel transport	Not tested	—	No extrapolation permitted
Restricted Moving-Fibre F16 ordering	Confirmed	14/16 eligible; natural gradient wins at all four radii in 14/14; wrong metric 0/14	Six algorithms, CNN–MNIST
Restricted action and realizability cost induce the same path ordering	Refuted	Natural-gradient cost ratio median 2.25, above 1.5 in 14/14; the same holds for pure moving-Fisher length, so the separation is not produced by the capacity factor	TinyCNN/MNIST, A_6 ; no combined scalar law is determined
A task residual is a precise address	Not supported	Substituting other tasks’ residuals costs little; effect gates failed	Artificial label tasks
One correction transfers universally across architectures	Not supported	Cosine and effect gates failed in both architectures	CNN / ViT
Ordinary learning selects the globally minimal path	Unsupported	Not tested; the audits compare constructed online realizations	Open
Full-backbone cross-architecture moving fibre	Open	Not tested	Backbone never adapted end to end

Claim	Status	Decisive evidence	Boundary
The complete response kernel forms a smooth bundle	Untested	—	Open

Frozen exclusions and their diagnostics are reported in Sections 5 and 7 and in Supplementary Section SA.

Full failed, excluded and diagnostically ineligible stages — the early cross-architecture attempts, the developmental and diagnostic Fisher cohorts, statewise dictionary transport, the temporal-selection and local-variation failures, and the GPT-2 distance audits — are retained with their values in Supplementary Section SA. Inadmissibility is evidence neither against a hypothesis nor for it.

10 Discussion: from parameter storage to fibre navigation

Three layers of “navigation” must be kept apart. *Confirmed*: an online transported chart can use current and past information to maintain $\text{im}B_k \subseteq \ker DR(\theta_k)$ throughout a path. *Supported as interpretation*: functional freedom is better represented as a state-dependent family than as a source-fixed null space. *Not shown*: that ordinary SGD, Adam or large-model training generally follows a globally minimal action. Navigation here is an operational property of causally generated current-kernel paths, not a theorem that unconstrained neural training naturally follows a horizontal lift or a globally minimal trajectory.

The evidence ladder is summarized below: selected null directions can carry adaptive value, the tested response-preserving chart moves, stale replay is costly, and minute transverse departures can be strongly amplified. On the mechanism side the layered picture is short: a task-sensitive base direction from source-local response geometry, a reusable neighbourhood from a correction learned across development tasks, and only a weak final ordering from the task’s own descriptor.

task-conditioned local geometry $\rightarrow$ base ray; shared learned correction $\rightarrow$ functional neighbourhood; own residual $\rightarrow$ weak final ranking

Mechanism layer	Judgment	Permitted statement
Local response geometry	Retained	Primary task-sensitive base direction
Shared correction family	Partial cross-family support	Coarse reusable prior; architecture dependent
Exact shared direction	Not supported	Cosine and effect gates fail
Precise residual address	Not supported	Substitution effects remain small
Natural prompt semantics	Not tested	No claim
Global argmin of the ledger	Not tested	Remains conjectural
Moving-fibre cost scaling	Confirmed in CNN and ResNet constructions	Current transport and source-fixed replay have distinct finite-radius scaling
Transverse response amplification	Confirmed in trainable ResNet layer4 + fc	Near-collinear directions can be functionally separated by the current response geometry
Complete response-kernel bundle	Not tested	No full-kernel or smooth-bundle claim
Restricted Fisher-action ordering	Confirmed, secondary	Finite six-algorithm, four-radius ordering of a declared action; capacity not isolated
Global CNER-F / CNER-S	Open	No arbitrary-path or universal claim

The evidence supports a third view of neural adaptation: relevant structure resides neither solely in individual parameters nor solely in the update rule, but in a state-dependent family of response-preserving directions navigated during learning. This reframes overparameterization without treating the entire kernel as functional: random controls show that adaptive value is concentrated in a comparatively small, geometrically identified part of a large null space, and identifying that part prospectively remains open.

10.1 Variational outlook: a constrained open problem

The confirmed result is a moving-fibre mechanism, not a global variational law. A global principle would quantify over a class of causal, realizable paths, but Sections 4–5 show that this class cannot be represented by a fixed source kernel: realizability depends on a response fibre that changes with state. The restricted F16 audit adds a second constraint. Moving-Fisher action and Newton-retraction cost select different paths, so neither quantity can be substituted for the other or combined with a fitted coefficient after the fact.

Any continuum theory must therefore derive its admissible path class and its objective independently. Plausible directions include a genuinely multi-objective Pareto formulation, a renormalized realization-cost density, or a higher-order term controlling connection, transverse sensitivity, or curvature. The amplification result makes the last possibility concrete: a small angular drift can enter a high-gain response-normal direction even when the two parameter directions remain nearly collinear. No such term is inserted into the present action, and no coefficient is inferred from the observed ordering.

The order of inference matters. Moving-fibre geometry determines which online proposals are response-preserving at the current state; only after that realizable class is specified does global path variation become meaningful. The present data therefore constrain future theory without repairing the declared F16 action after the fact.

11 Limitations

11.1 Model scope

Each confirmed claim carries its own model scope, and they are not interchangeable. Selected tangent value is confirmed in the complete final-classifier parameter space of a frozen ImageNet-pretrained ResNet-18 representation, across three anchor counts and sixteen seeds, in addition to the original CNN and ViT audits. Moving-fibre cost scaling is confirmed in the CNN-MNIST construction and in a frozen-backbone adapter-plus-classifier ResNet construction. Transverse response amplification is confirmed in a trainable layer4-plus-classifier system. A frozen GPT-2 rank-2 LoRA domain adds cross-domain evidence: selected tangent value is confirmed on codeword prompts, while current-versus-fixed cost separation is prospectively confirmed on disjoint natural-English development and response continuations. The restricted F16 ordering and the action-cost separation remain confined to TinyCNN/MNIST. The sixteen seeds in each ResNet cohort vary training, not the dataset, pretrained representation or architecture.

11.2 Geometric scope

The transported chart is eight-dimensional throughout: a controlled subchart of the response kernel, not the complete kernel, whose numerical dimension in the CNN construction is 4146. The complete ResNet backbone is never adapted, the GPT-2 audits are confined to a rank-2 LoRA-B block of the last two layers rather than the complete parameter space, no global smooth fibre bundle is established, and nothing is claimed across rank-changing or singular regions. Natural language-model training trajectories are not tested. On a constant-rank region $\ker DR$ is already the level-set tangent distribution, so its local

integrability is not what is at issue. The capacity H_{cap} is declared rather than derived, and its sensitivity to R , Q , the ridge, the capacity floor and the probe set has not been mapped.

One geometric limit concerns the unit in which every effect in Section 3 is measured. Since $\dim \ker DR(\theta) = N - \text{rank } DR(\theta)$, the fraction of any realized update that is response-preserving is set by the declared anchor set, a free design parameter. A small anchor set leaves most of every update inside the fibre; a large one leaves little there. No sweep over anchor count, kernel dimension or realized tangent share accompanies those results, so the defensible form of the finding is the ordering, not the magnitude.

11.3 Variational scope

The action ordering is an argmin over the frozen six-algorithm family A_6 at four finite radii. It is not a statement about arbitrary causal paths, an Euler–Lagrange condition, or a global minimality theorem. The arms share a functional but not a parameter endpoint; the Newton-retraction Fisher length is an operational proxy, not the unique notion of realizability cost; four radii do not establish an exact continuum limit; the leading coefficient or renormalized density of the vanishing realizability cost remains uncertified; and no optimizer is shown to select the current-kernel transported chart.

11.4 Statistical and protocol scope

The confirmations have different margins and should not be summarized by the tightest of them. The cost-scaling cohort passed 15/16; the F16 cohort passed 14/16, exactly meeting its gate and leaving no eligibility margin; the transverse cohort passed 16/16 against a frozen 12/16. Two Fisher cohorts remain formally ineligible and are recorded as such in Supplementary Section SA. The Fisher metric may change with chart basis, sampling distribution, damping and response definition. The dictionary underlying Section 3 is finite and demonstrably incomplete, several diagnostic arms use oracle tangent norms, the low-dimensional geometric audits use three-parameter models in which many endpoint-matched paths are close by construction, and six regimes are too few for stable population inference, so $p = 0.21875$ must be read as reported. Separate GPT-2 soft-prompt distance audits in Supplementary Section SA test source-frozen Fisher length only and make no result here cross-architecture.

11.5 Next steps

The verification rungs that follow, in order: complete layer4 and then full-backbone chart transport; a sweep over anchor count and chart dimension, so that reported magnitudes become a function of the realized tangent share rather than a single operating point; replication of transverse amplification in another modern architecture and data domain; a Moving-Fibre F16 ordering on ResNet; dense local path variation; and only then a global variational treatment. The full milestone programme, with each milestone’s primary gate and the meaning of its failure, is in the Supplementary Prospective Programme.

12 Conclusion

Selected response-preserving directions are not inert redundancy. Their adaptive-value ordering survives prospective confirmation in the complete final-classifier space of a frozen ResNet-18 representation, with all sixteen seeds passing across three anchor counts. More importantly, the CNN and ResNet audits establish the central moving-fibre mechanism: transporting a direction in the current response kernel and replaying a direction frozen at the source produce systematically different finite realizability-cost scaling. In CNN–MNIST, all fifteen eligible seeds passed, with $\alpha_{\text{moving}} = 0.684$ against $\alpha_{\text{fixed}} = 0.007$ and a median cost ratio of 21.30; the independent ResNet-18/CIFAR-10 confirmation passed 16/16 seeds.

A trainable ResNet block further shows why this distinction matters: current-fibre and source-fixed directions remain nearly collinear, yet their active response residuals differ by more than three orders of magnitude. The relevant object is therefore not merely a large null space, but a moving null distribution embedded in a strongly anisotropic response geometry.

The restricted action audit also fixes the present theoretical boundary. The constructed true-natural-gradient path minimizes the declared Moving-Fibre F16 action among six causal online algorithms at every tested radius, but it is systematically not the least expensive path to realize, paying a median 2.25-fold correction-cost premium. Thus moving-fibre geometry, path selection and path realization cannot yet be identified with a single global variational functional. The result constrains any future action principle rather than proving one.

GPT-2 provides cross-modal confirmation of this CNN/ResNet mechanism rather than a second independent centre of the paper. A rank-2 LoRA-B audit first established selected tangent value and reproduced the current-versus-fixed separation on constructed codeword paths. The decisive prospective extension used disjoint frozen natural-English development and response continuations: all 8/8 new seeds and 32/32 interior nodes passed, with $\alpha_{\text{current}} = 1.9987\text{--}2.0040$, $\alpha_{\text{fixed}} = 1.0625\text{--}1.2106$, and finest-radius fixed-to-current correction-cost ratios of 8.87–35.56. This rules out confinement of the observed separation to the original codeword prompts within the declared experiment. It does not establish semantic invariance, full-parameter GPT-2 geometry, ordinary training navigation or a language-model variational principle.

The strongest present conclusion is therefore a structural separation. Selected adaptive freedom resides in a state-dependent response family; finite realization depends on transporting that family with the state; and path action and realization cost remain empirically non-equivalent. CNN and ResNet establish this mechanism and its geometric boundary, while GPT-2 confirms that the separation extends to a different architecture and modality within a declared low-rank subspace. Whether a derived global principle can unify these quantities remains open.

Appendices

Method detail and the full development record are held outside the main text so that the argument chain above stays legible.

Appendix	Contents	Location
A	Frozen datasets and architectures	Supplementary Information
B	Jacobians, metrics and projections, including the coordinate-covariance results for the capacity construction	Supplementary Section SB
C	Prospective protocol and the full gate thresholds for every stage	Supplementary Information
D	Full tangent-value mechanism tables: dictionary reuse, statewise transport, residual address factorization and the shared-correction controls	Supplementary Tables S2–S6
E	Moving and fixed arm seed-level results with exponent fits and residuals, and the complete v3.2c six-algorithm, four-radius run record	Supplementary Item 13 and the public archive
F	Negative and ineligible experiments, including the developmental and diagnostic Fisher cohorts and the GPT-2 distance audits	Supplementary Sections SA–SB
G	v4.1b sixteen-seed cross-model cost-scaling confirmation	Pending release
H	v4.2c precision failure and v4.2c-r1 float64 repair	Pending release
I	v4.2d sixteen-seed prospective transverse-amplification confirmation, with per-seed summaries, every gate, ranges rather than medians alone, audited path-state counts, nonzero transverse states, finite/JVP radius convergence, JVP additivity and the TF32/float64 environment	Pending release

Data and code availability

The public archive is at <https://github.com/papasop/neural-fibre-geometry>. The records for the CNN–MNIST moving-fibre and F16 lines and for the frozen-classifier ResNet confirmation are archived at commit 698e99fd9c78d765011db8e8f32876903c30bf37, release tag v1.3.0-resnet-external-confirmation. That archive contains, for each confirmed cohort, the frozen protocol and its identifier, per-seed JSON records, the combined result and report, chart and metric certifications, integration-convergence records, exclusion records and the original result ZIPs, together with the failed, developmental and ineligible stages.

Artifact	SHA-256
v3.1e engine	00e4ade63c441a9d4a83e814177a74b6f1414ff792adb7e9601a80a0b900afbd
v3.2c engine	8b3ba6f37bc57bad5b3752238831366c668f94bf7959c2f45cf5749d016cbb52
v4.0c-r1 engine	dca49cb187c59a2ac8e3e55fe4970349f258fed36b7ffbfca510770c19af9f63
v4.0c-r1 raw results	a3edf61ed8fcd029f302f922650071dec9a1b60c501ace34a4ec6965ebf71823

Supplementary Item 13 provides the complete v3.2c run record: SI_V32C_PATHS_384.csv contains all 384 algorithm–radius paths, SI_V32C_SEED_GATES_16.csv the eligibility and frozen-gate outcomes for all sixteen seeds, and SI_V32C_EXCLUSIONS_2.csv the exact failed statistic and regular diagnostics for seeds 73730 and 73732. Post-confirmation radius leave-one-out outputs and their reconstruction script are under docs/robustness_v3_1e/ and audits/formula_audit_v1/.

The v4.1b cross-model scaling confirmation, the v4.2c precision failure, its float64 repair, the v4.2d transverse-amplification confirmation, and the GPT-2 LoRA r3, v2 and natural-text v3 audits with their earlier r1 and r2 failure records are not yet in the public archive. Their per-seed records, frozen protocols, gate sets, run environments and engine hashes will be released, with the failure-to-repair-to-confirmation history intact, in a subsequent tagged release. Sections 3.5, 5.3, 5.4 and 6 should be read against that pending release rather than against the commit above.

Summary tables alone are not sufficient: the present claim rests on the eligibility judgement of fourteen complete six-algorithm, four-radius records — twenty-four arms per seed — which a referee must be able to re-check path by path. Per-seed records for the Section 3 mechanism audits are not yet in the archive, and the pooled tests that depended on them are withdrawn rather than reported.

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